A new attack surface that exploit runtime behaviour on different kernels, with potential to inspire follow-up research.

Official Review by Reviewer NFCY 12 Jan 2026, 22:39 (modified: 21 Feb 2026, 14:45) Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer NFCY, Authors Revisions

Paper Summary:
The paper observes that the same DNN model may exhibit different runtime behavior (i.e., inference outputs) when executed on different kernels, for example due to differences in optimization mechanisms. It argues that this discrepancy introduces a new attack surface: a maliciously constructed model M could behave benignly when executed on one kernel K1, yet exhibit malicious behavior when run on another kernel M2. To support this claim, the paper provides a concrete construction that demonstrates such an attack.

Under the proposed threat model, the attacker is assumed to have full control over the training and construction of the model M, , and has "black-box" access to both kernels K1 and K2 (i.e. the attacker can execute arbitrary models on the kernels and observe their outputs.)

Paper Strengths:
The proposed attack surface is new and practical. It considers the runtime and inference behavior of the underlying machine learning kernels, which have not been previously discussed.

Major Weaknesses:
There could be easier construction, since the attacker has full control on the construction of the DNN model. For example, the attacker could simply insert a component that detects the kernel, and then conditionally outputs the desired result. The resulting model could then be obfuscated prior to release to reduce the likelihood of detection.

Minor Weaknesses:
The abstract characterizes the proposed method as a "red-teaming attack," yet the justification for this terminology is unclear. The paper seems to present an attack demonstration rather than a comprehensive red-teaming exercise. In established cybersecurity practice, red-teaming generally encompasses additional components such as defensive evaluation, remediation, and follow-up analysis.

Preliminary Recommendation: 5: Weak Accept
Justification For Recommendation And Suggestions For Rebuttal:
The proposed attack surface is novel. While it may appear obvious once explained, and the techniques used to realize it are unsurprising, the idea itself is innovative and non-trivial.

This work has the potential to inspire follow-up research.

Confidence Level: 4: High Confidence - The reviewer has strong expertise in the area. They are highly familiar with the relevant literature and can critically evaluate the paper.

New System-Aware Attack Surface, but Optimization Details and Metrics Lack Rigor

Official Review by Reviewer X5Mw 12 Jan 2026, 16:52 (modified: 23 Jan 2026, 02:43) Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer X5Mw, Authors Revisions

Paper Summary:
This paper introduces VLMSysTrojan, a system-aware backdoor attack for VLMs. It exploits floating-point non-associativity differences between cloud kernels (training) and edge kernels (deployment). By optimizing a trigger and a guard-bias, the model is designed to behave benignly on cloud systems but activate a backdoor on target edge systems.

Paper Strengths:
1. This paper exploits system-level kernel discrepancies (floating-point errors) rather than model weights or data poisoning alone.
2. The proposed backdoor can bypass pre-deployment inspections since the backdoor is inactive on standard cloud kernels.
3. The authors evaluate the method across multiple VLM architectures (BLIP, BLIP-2, LLaVA-1.5) and datasets, validating the feasibility of the proposed mechanism.

Major Weaknesses:
1. The definition of Stealthiness Retention Rate (SRR) is inconsistent. While the mathematical definition (referenced as Eq. 9) suggests it measures behavior on triggered inputs (to ensure dormancy), the text (Lines 394-407) claims it assesses behavior on benign inputs.
2. The SRR metric assumes the ability to verify whether the trigger activates the backdoor. However, in a real-world defense scenario, the "stealthiness" is difficult to quantify because the defender does not know what the trigger looks like. The paper relies on the prior knowledge of the specific backdoor trigger to calculate SRR. This makes SRR a theoretical metric for the attacker rather than a practical reflection of real-world detectability.
3. It is unclear how Equation 4 is implemented in practice. The objective function requires minimizing loss across two different kernel systems K_cloud and K_target simultaneously to optimize the parameters θ. How are gradients propagated back to the model parameters θ during the training phase? The paper lacks technical details on how this cross-kernel optimization is achieved.
4. The mechanism relies on precision differences. The authors should explicitly discuss the distinction between this kernel-level attack and existing quantization-based backdoors[1].
5. It is unclear if Eq. 4 optimizes the visual encoder or only the decoder. If the visual encoder is updated, it would likely invalidate the trigger r_* and guard-bias β optimized in previous steps (Eq. 2), reducing attack effectiveness.
6. The ablation study shows that using only "Module 3" (Decoder Finetuning) causes accuracy (Pre/Post-Func) to drop to ~25%. Since this is essentially threshold-based fine-tuning, such a drastic performance collapse on benign data is counter-intuitive and unexplained.

[1] Ma H, Qiu H, Gao Y, et al. Quantization backdoors to deep learning commercial frameworks[J]. IEEE Transactions on Dependable and Secure Computing, 2023, 21(3): 1155-1172.

Minor Weaknesses:
1. There is a typo in Eq. 3. The condition "M_θ^{vis}(x + r, K_cloud) > β" should logically be defined on "K_target", as the goal is to activate the backdoor only on the target kernel.

Preliminary Recommendation: 2: Weak Reject
Justification For Recommendation And Suggestions For Rebuttal:
Although the idea of leveraging system-level floating-point discrepancies is novel, the paper suffers from significant ambiguities in the description of the optimization methodology and the definitions of key metrics. The authors should provide clear and precise clarifications on these points in their rebuttal.

Confidence Level: 4: High Confidence - The reviewer has strong expertise in the area. They are highly familiar with the relevant literature and can critically evaluate the paper.

Review of Submission 37232

Official Review by Reviewer tnYg 10 Jan 2026, 17:28 (modified: 21 Feb 2026, 14:45) Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer tnYg, Authors Revisions

Paper Summary:
This paper presents VLMSysTrojan, a system-level backdoor attack targeting vision-language models. The attack works by optimizing a trigger to amplify the floating-point inconsistencies across different kernel systems, and a "guard-bias" is computed to ensure that the triggered behavior only happens when the input is triggered and applied on the target system. Extensive experiments demonstrate that the proposed method is effective.

Paper Strengths:
- The idea of exploring "system-level" backdoors for VLMs is interesting. Compared to traditional "model-level" backdoor attacks, this method is more realistic and can encourage researchers to reconsider the security of current foundation models from broader perspectives.
- The paper is in general well-written and easy to follow.
- Extensive experiments demonstrate that the proposed method is the only method that can achieve high stealthiness while maintaining comparable clean functionality compared to the baselines.

Major Weaknesses:
- The threat model is a bit strong. In detail, the attacker is assumed to "run the model on both the cloud kernel environment and the target deployment kernel environment." If this information is unknown, is it possible the proposed attack can transfer?
- Compared with traditional backdoor, the proposed method involves trigger optimization, guard-bias grid search, and decoder fine-tuning, appearing significantly more complex than the baselines. It would be beneficial for the authors to provide a comparison of the computational resources required.
- The trigger optimization explicitly exploits floating-point inconsistencies. However, real-world deployments often utilize integer quantization to improve inference speed. Consequently, the paper lacks an evaluation of robustness against quantization, which is critical for practical applicability.

Minor Weaknesses:
N/A

Preliminary Recommendation: 4: Borderline Accept
Justification For Recommendation And Suggestions For Rebuttal:
New attack, but may not be realistic in real-world scenarios.

Confidence Level: 3: Moderate Confidence - The reviewer is reasonably knowledgeable about the topic. They understand the paper's methodology and results but may not be a leading expert in the specific subfield.